Performance based Algorithm for the Detection and Extraction of Human Skin

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***Abstract*–This paper presents the comparative analysis ofvarious color space models like RGB, HSV and YC_bC_rcolor models for detection and extraction of human skin from it.Skin detection is broadly used in applications like face detection, tracking of the body parts and hand gesture analysis, for retrieval of skin and blocking ofunpleasant contents. The proposed method is performed in four phases;the effect of color space transformation on skin detection performance and finding the appropriate color space for skin detection; the role of the illuminancepart of a color space; the appropriate pixel based skin color modeling technique; and finally, the effect of color fidelity algorithms on color based skin classification. The heterogeneityof RGB color space isdetached by using HSV and YC_bC_rcolor space models.In this paper the input image captured from a camera is converted into three different color space models i.e. RGB, HSV and YC_bC_r and then performance analysis of algorithms is to be evaluated on the basis of skin detection and extraction.**

***Keywords*- skin detection, color space transformation,RGB, HSV, YC_bC_r**

I. INTRODUCTION

Skin detection can be defined as the method of selecting which pixels of a given image match to human skin. Skin detection is helpful inface detection and face tracking for security and video indexing applications, model-based video coding, etc. [1, 2]. There are some difficultieswhile detecting skin pixels [3]. Skin color is affectedby ambient light which is unknown in many situations; different cameras produce non uniformcolors, even for the same person, under the same ambient conditions of lighting; and also, skin colorsalter from person to person. Numerous color space models have been discussed in the literature for skin detection applications. YC_bC_r has been broadly used for the skin pixels from a compressedbunch in the C_b-C_r plane. YC_bC_rcan also be used in video coding and therefore no moreconversion is required, this color space model has been used in

skin detection applications where the video series is compressed. Skin and face detection using an automatic detection tool or software plays very important roles in many visualization applications, such as recognition of human face and gesture in visual surveillance and intelligent human-machine intelligence [4-6].

Most existing skin segmentation techniques involve the classification of individual image pixels into the skin and nonskin categories on the basis of pixel colors. The inspiration behind this approach is that the human skin has very consistentcolors which are different from the colors of many other things. In the earlier years, a number of comparative studies of skin color pixel classification have been reported. Jones and Rehg [4] created the first large skin database—the Compaq database—and used the Bayesian classifier with the histogram technique for skin detection. Brand and Mason [5] compared three different techniques to the Compaq database: thresholding the ratio of redcomponent togreen component, color space mapping with 1D indicator, and RGB skin possibility map, while Terrillon*etal*. explained comparison between Gaussian and Gaussian mixture models across many chrominance places with a set of 110 images of30 Asian and Caucasian persons[6]. Also Shin *eta*l. [7] compared skin segmentation in various color spaces. In their study, skin samples and non skin sampleswere taken from the two different databases.

In this paper, the authors were attempting to presentthe performancebased algorithm for skin detection on different color space models (i.e.RGB, HSV and YC_bC_rcolor space model)for real time imagescaptured by the camera

II. COLOR SPACE MODELS

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In the past, different color spaces have been used in skinsegmentation. In some cases, categorization of color is done by using only pixel chrominance because it is expected that skin segmentation may become more robust to lighting variations if pixel luminance is discarded. In this paper, we investigate how thechoice of color space and the use of the chrominance channels affectskin segmentation. We should note that there are presentplentiful colorspaces but many of them share similarcharacteristics. Therefore, inthis paper, we focus on three colors spaces which arecommonly used in the image processing field.

a. RGB Color Space

Since RGB as shown in Figure 1 does not decouple the effects of luminance. It is still able to eliminate certain colors that are clearly out of the range of the normal skin color. From different study and experiments with various thresholds in RGB color space, we found that the subsequent rule worked well in removing some unnecessary pixels. [8]

Range in Skin area for RGB color space is:

$$0.836G-14<B<0.836G+44 \rightarrow [skin]$$

and

$$0.79G-67<B<0.7.8G+42 \rightarrow [skin]$$

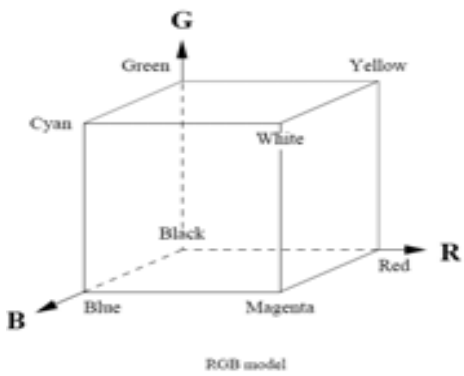

Figure 1:RGB color space system

b. HSV Color Space

As known RGB is the most commonly used basis for color descriptions, it has the negative characteristic that each of the coordinates (red, green, and blue) have different values and also sometimes it does not necessarily provide relevant information about whether a particular image "patch" is skin or not a skin however it is subject to luminance effect from the lighting intensity of the surroundings. The HSV color space model is shown in Figure 2 provides color information in a manner more in line how humans think of colors and how artists typically mix colors. "Hue" describes the basic pure color of the image, "saturation" gives the manner by which this pure color (hue) is diluted by white light, and "Value" provides an achromatic notion of the intensity of the color. It is the first two, *H* and *S* that will provide with useful discriminating information regarding skin [9, 10].

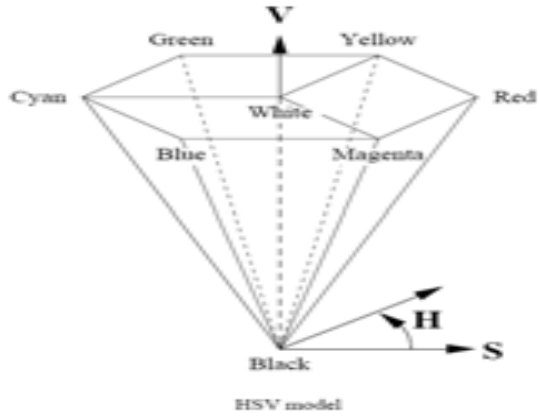

Figure 2: HSV color space system

This is the most noticeable trend used to derive the following rule is:

$$HD_i = \max(H_i, S_i, V_i)$$

Where,

$$H_i = \sum_{b=1}^{D} \left| h_i^H(b) - h_{i+j}^H(b) \right| \quad \ldots(1)$$

$$S_i = \sum_{b=1}^{D} \left| h_i^S(b) - h_{i+j}^S(b) \right| \quad \ldots(2)$$

$$V_i = \sum_{b=1}^{D} \left| h_i^V(b) - h_{i+j}^V(b) \right| \quad \ldots(3)$$

c. YC_bC_rColor Space

After analyzing over wide range of threshold the best performance is found by using matrix given below on YC_bC_r. It isalso well proved that YC_bC_rcolor space is more efficient tool for the elimination of non skin area than the other color space models.And after experimenting with various thresholds, the best results were found by using the following rule [11-12].

$$\begin{pmatrix} Y \\ Cb \\ Cr \end{pmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168636 & 0.232932 & -0.064296 \\ 0.499813 & -0.418531 & -0.081282 \end{bmatrix}$$

Figure3 shows the schematic of YC_bC_rcolor space system

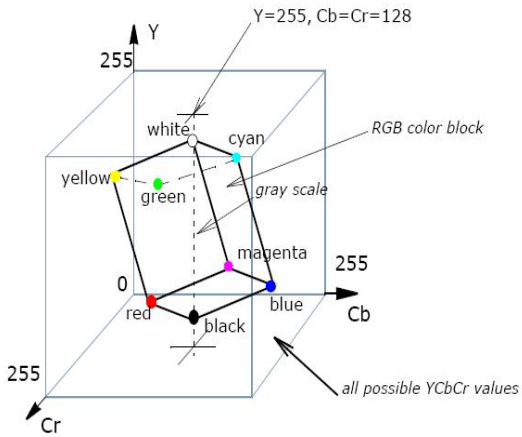

Figure 3: YC_bC_r color space system

Range for Skin Region:

$$102 < C_b < 135 \rightarrow \text{Skin}$$

and

$$135 < C_r < 178 \rightarrow \text{Skin}$$

III. PROPOSED ALGORITHM FOR HUMAN SKIN DETECTION

The Input is RGB image which has been captured from Web Cam which consists of three layers red, green and blue.The input images are captured by web camera as programmed by the algorithm. Some of the samples are shown in figure 4.

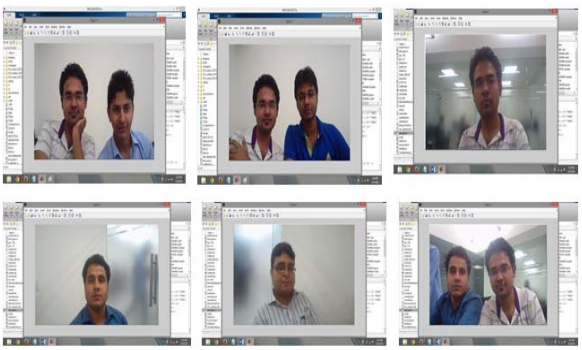

Figure 4: Sample RGB image captured from web camera

a. Segmentation and Detection of Human Skin using RGB

The RGB image captured is processed using the proposed algorithm for the detection of human skin by applying the threshold on R, G and B components and obtained the segmented image. [12, 13]

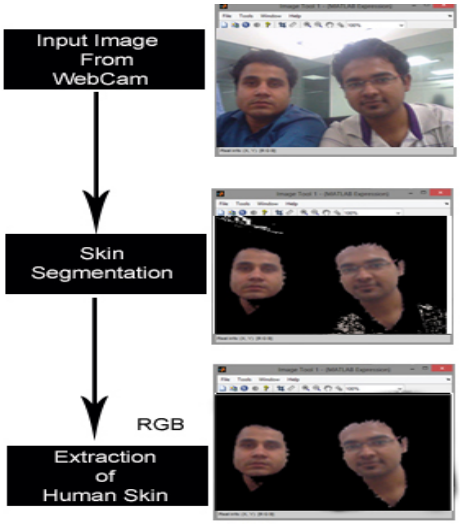

Figure 5: Segmentation and extraction of image using RGB Colour Space Model

b. Conversion from RGB Image to HSV Image

To convert the RGB image into HSV image separate the hue component *H*. *H* component represents the basic pure color of the image, while *S* and *V* component represents the luminance information which has more variation and that is the reason these components are discarded. By applying the threshold on hue component the segmented is obtained which has been shown in figure 6. [14]

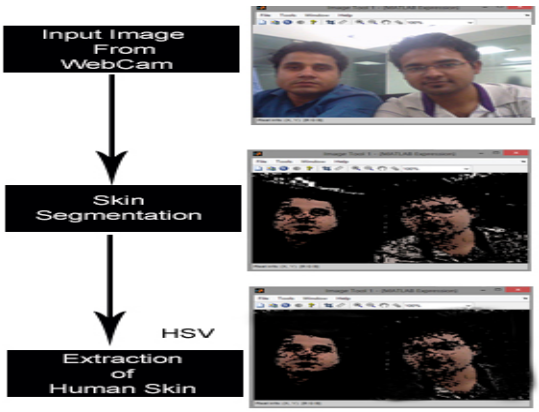

Figure 6: Segmentation and extraction of image using HSV Colour Space Model

c. Conversion from RGB Image to YC_bC_rImage.

To convert the RGB image into YC_bC_rimage, separate the chroma component $C_b$and C_r. Y component represents the luminance information which has more variation that is why discard this component, C_b and C_r component are used. Apply the threshold on these components and obtained the segmented image which is shown in figure 7.[14]

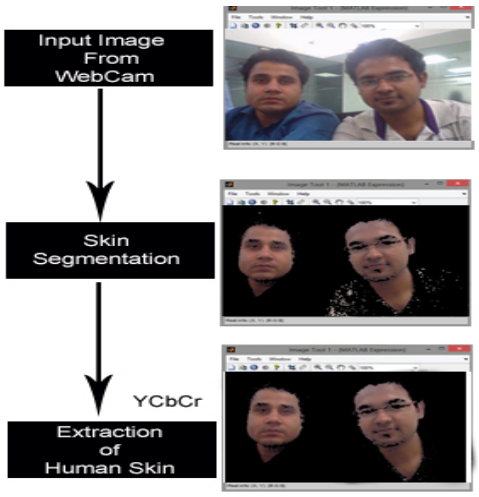

Figure7: Segmentation and extraction of image using YC_bC_rColor Space Model

IV. RESULTS

Analyzing skin color segmentation with all three color space models in real time, obtained results are shown in figure 8 for all color space models.

(a) Human Skin Detection using RGB color space

(b) Human Skin Detection using HSV color space

(c) Human Skin Detection using YC_bC_r color space

Figure 8:Extraction of human skin with RGB, HSV and YC_bC_rcolor space

V. Comparison between RGB, HSV andYC_bC_r Color Space

This section presents detailed experimental comparison of RGB, HSV and YC_bC_ralgorithms. Comparison of skin color segmentation with three color model shows that the performance of YC_bC_r is better than the rest of the color space model.HSV color space has certain advantages over RGB when it comes to face detection since the skin colours are sensitive to the lighting condition, RGB is a concise system for describing colors, and it is somewhat difficult for us to describe the nature and amount of a color channelized for eye.In Contrast to RGB, the YC_bC_rcolor space is independent of luminance and thus it gives better performance.

Table 1: Performance comparison between RGB, HSV and YC_bC_r

S. No.	Algorithm	Sample Images	Image Detection	Performance
1.	RGB	40	30	85.00%
2.	HSV	40	32	92.50%
3.	YC_bC_r	40	38	95.50%

VI. CONCLUSION

YC_bC_rcolor space is an absolute color space, it defines colorsexactly. YC_bC_r is better among three color systems for it gives more image information. We face many problems duringsegmentation due to different light

condition
and background color. After performing these segmentation methods weconclude that YC_bC_r is good for skin color segmentation.

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